

ABSTRACT

The field project titled “**Survey on E-Waste Awareness and Recycling Habits**” investigates the knowledge, attitudes, and practices of individuals regarding electronic waste (e-waste). With rapid technological advancement and shorter product life cycles, e-waste has emerged as one of the fastest-growing waste streams, creating environmental and health hazards due to the presence of toxic substances. The study employed a **mixed-method approach**, with primary data collected through a **Google Form questionnaire** distributed via online and offline modes, supplemented by informal interviews. A total of **51 responses** were obtained from teachers, family members, and friends, and analyzed using **Google Forms and Microsoft Excel**. The questionnaire captured demographic details, awareness levels, disposal behaviors, motivators, and barriers related to e-waste management. The findings indicate that while most respondents were aware of the harmful effects of e-waste, knowledge about **government rules, authorized recycling centers, and safe disposal practices** remained limited. Instead, common practices included **storing or selling devices to scrap dealers** rather than formal recycling. The study concludes that although awareness exists, there is a significant gap between knowledge and practice. The results emphasize the urgent need for **awareness campaigns, accessible collection facilities, incentives for recycling, and stronger enforcement of e-waste management rules** to promote sustainable practices.

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Figure 1: Collaborative use of Google Forms for digital e-waste survey data entry.

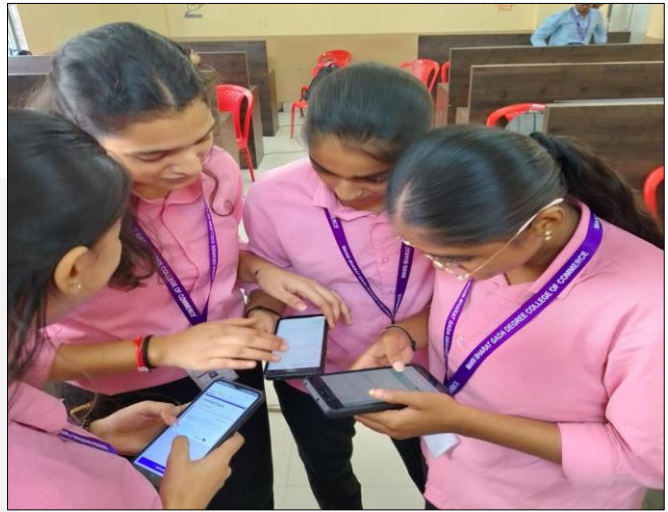


Figure 2: Mobile phone utilized for Google Form response input.



Figure 3: Students engaging in the e-waste survey using mobile devices.



Figure 4: Survey participant responding to e-waste awareness questions at home.

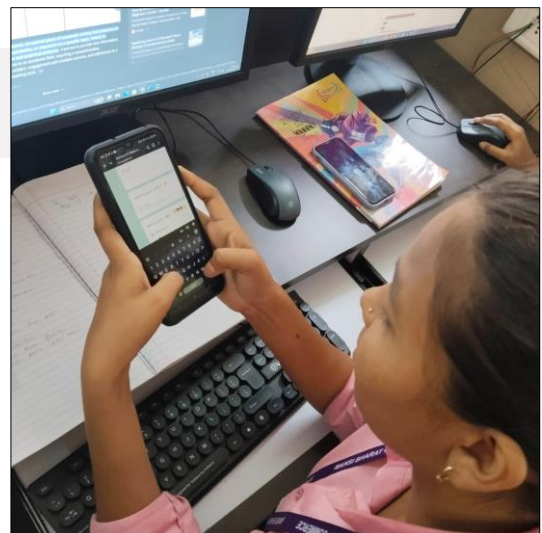


Figure 5: Data collection through Google Form during field project work.

CHAPTER 1: INTRODUCTION

In the modern era, technology has become an essential part of everyday life. Devices such as **mobile phones**, **computers**, and **household appliances** have improved communication, education, and comfort. However, the increasing use of these devices has also created a major concern in the form of **electronic waste (e-waste)**.

E-waste refers to discarded electronic products that are no longer useful. Unlike ordinary waste, it contains harmful substances such as **lead**, **mercury**, and **cadmium**, which can pollute the environment and pose risks to **human health** if not managed properly.

The issue has grown rapidly due to **shorter product life cycles** and people's demand for the **latest technology**. Many individuals are unaware of **safe disposal methods** and often **dump** or **burn e-waste**. At the same time, e-waste holds valuable materials like **copper** and **gold** that can be recovered through **proper recycling**.

This makes **awareness** and **recycling habits** extremely important. **Educational institutions**, **local communities**, and **government authorities** play a key role in spreading awareness. **Field projects** and **surveys** provide insights that can guide better **policies** and promote **responsible recycling** in the future.

CHAPTER 2: LITERATURE REVIEW

Since 2025 :

- [Assessing the relationship of **knowledge** and attitudes on **e-waste recycling** practices among students of higher learning institutions](#)

✚ NS Yaacob, [NANN Azman](#)... - Journal of Nusantara ..., 2025 - journal.unisza.edu.my

Since 2024 :

- [**Awareness, Safety Practices and Associated Factors Among E-Waste Recycling Workers in Bangladesh**](#)

✚ MN Munni, [MR Karim](#), [M Haque](#)... - Environmental ..., 2024 - journals.sagepub.com

Since 2021 :

- [**E-waste recycling behaviour: An integration of recycling habits into the theory of planned behaviour**](#)

✚ [M Aboelmaged](#) - Journal of Cleaner Production, 2021 - Elsevier

Sort by relevance :

- [**Understanding the individuals' motivators and barriers of e-waste recycling: A mixed-method approach**](#)

✚ [R Yadav](#), [DK Panda](#), S Kumar - Journal of Environmental Management, 2022 – Elsevier

Review Articles :

- [**Consumer behaviour towards storage, disposal and recycling of e-waste: systematic review and future research prospects**](#)

✚ [S Thukral](#), [D Shree](#), [S Singhal](#) - Benchmarking: An International ..., 2023 - emerald.com

CHAPTER 3: METHODOLOGY

3.1 Research Method :

The study followed a **Mixed Method Research approach**, using quantitative questions (MCQs, checkboxes) to measure awareness and recycling habits, along with qualitative short answers (names, emails) for authenticity.

3.2 Tool Used :

A **Google Form questionnaire** (software/app) was the main tool, supported by informal **interviews** with teachers, family, and friends for additional clarity.

3.3 Sample Size :

A total of **51** responses were collected from **teachers, family, and friends**, covering different age groups and backgrounds.

3.4 Sampling Method :

The study used **Non-Probability Convenience Sampling**, with elements of **Snowball Sampling** as some participants shared the form further.

CHAPTER 4: FIELD WORK DESCRIPTIONS, OBSERVATION AND ANALYSIS

4.1 Field Work Descriptions :

For my field project on **E-Waste Awareness and Recycling Habits**, I collected data mainly through a Google Form shared with teachers, friends, and family, both online and offline. The form was kept simple for easy responses, and I also had small conversations with a few participants. This fieldwork gave me useful experience in reaching people and understanding their views on e-waste.

4.2 Observation :

It was observed that while many respondents were aware of the harmful effects of e-waste, proper knowledge of safe disposal and recycling methods was still lacking. Most preferred storing or selling old devices rather than using authorized recycling centers, showing that awareness is higher than actual practice. Younger participants, especially students, showed slightly better awareness compared to others.

4.3 Analysis :

Maximum respondents correctly identified common electronic items as e-waste, and most knew about its harmful effects on health and the environment. Most did not know proper disposal methods or authorized recycling centers, and the majority were unaware of government rules. Maximum respondents disposed of e-waste at home or through scrap dealers, with few segregating it, and irregular disposal was common. Most preferred local collection centers, while awareness of recycling drives was limited, and few were motivated by incentives. Maximum respondents suggested more awareness campaigns and better accessibility to collection centers, whereas fewer suggested rewards for recycling or strict enforcement of e-waste laws.

CHAPTER 5: CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion :

The survey reveals that while respondents are generally aware of what constitutes e-waste and its harmful effects, there is a significant lack of knowledge regarding proper disposal methods, authorized recycling centers, and government regulations. Most e-waste is disposed of at home or through scrap dealers without segregation, reflecting irregular and unsafe practices. Although respondents show a preference for local collection centers, awareness of organized recycling drives and incentives is low. To improve e-waste management, there is a clear need for increased awareness campaigns, better accessibility to collection centers, and initiatives to encourage safe disposal practices.

5.1 Recommendations :

- 1. Increase Awareness Campaigns:** Organize regular programs in schools, colleges, and communities to educate people about e-waste hazards and proper disposal methods.
- 2. Improve Accessibility:** Set up more local e-waste collection centers and ensure they are easy to locate and use.
- 3. Provide Incentives:** Offer rewards or benefits to encourage people to recycle e-waste responsibly.
- 4. Promote Segregation Practices:** Encourage households and businesses to segregate e-waste from general waste before disposal.
- 5. Strengthen Regulations:** Implement and enforce laws regarding safe e-waste management to reduce improper disposal.
- 6. Support Recycling Drives:** Organize periodic recycling drives and collection camps to make recycling more convenient and visible.

REFERENCES

Primary Source

- Google Form Survey. (2025). *Survey on E-Waste Awareness and Recycling Habits* [Unpublished raw data]. Google Form link: <https://forms.gle/BCJCoDdztzkfB6nAA>

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- Central Pollution Control Board. (2023). *Annual report on e-waste management in India, 2022–23*. Government of India.
<http://cpcb.nic.in>
- International Telecommunication Union. (2020). *Global e-waste monitor 2020*. <https://www.itu.int>

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- Aboelmaged, M. (2021). E-waste recycling behaviour: An integration of recycling habits into the theory of planned behaviour. *Journal of Cleaner Production*, 278, 123956.
<https://doi.org/10.1016/j.jclepro.2020.123956>
- Yadav, R., Panda, D. K., & Kumar, S. (2022). Understanding individuals' motivators and barriers of e-waste recycling: A mixed-method approach. *Journal of Environmental Management*, 301, 113858. <https://doi.org/10.1016/j.jenvman.2021.113858>

AI Tool Used

- OpenAI. (2025). *ChatGPT (GPT-5)* [Large language model]. OpenAI. <https://chat.openai.com>

Appendix C – Sample Data (Excel Table)

Sr.no.	Timestamp	Email address	What Is your name ?
1.	17/09/2025 15:46:30	nus.gmm@gmail.com	Nusrat
2.	17/09/2025 16:11:30	rehangauri310@gmail.com	Rehan
3.	17/09/2025 16:19:24	snehagoli42@gmail.com	Goli sneha Naresh
4.	17/09/2025 16:25:48	deepakgavali231@gmail.com	Deepak gavali
5.	17/09/2025 16:30:32	siminkhan43@gmail.com	Simin khan
6.	17/09/2025 17:23:10	kaveriyamewarrollno@gmail.com	Kaveri Yamewar
7.	17/09/2025 17:23:22	ns9494964@gmail.com	Neha shah
8.	17/09/2025 17:29:02	anushkavikasthakare@gmail.com	Anushka Thakare
9.	17/09/2025 17:48:34	yelgamp@gmail.com	Yelgam Pallavi Pandu
10.	17/09/2025 17:49:26	ns9356621@gmail.com	Singh Akansha Shivsankar
11.	17/09/2025 17:58:25	amolkamble24766@gmail.com	Amol Kamble
12.	17/09/2025 18:17:01	ansariparveen06@gmail.com	Parveen ansari
13.	17/09/2025 18:17:47	arzoansari5866@gmail.com	Arzoo Ansari
14.	17/09/2025 18:20:41	guptarajveer1224@gmail.com	Rajveer
15.	17/09/2025 18:31:49	nirmitip1105@gmail.com	PATIL NIRMITI SACHIN
16.	17/09/2025 18:37:18	mominkainat05@gmail.com	Momin Kainat Mehmood Hasan
17.	17/09/2025 18:46:02	sonalimali421@gmail.com	Sonali Ashok Mali
18.	17/09/2025 18:49:11	shuklasneha8730@gmail.com	Shukla sneha
19.	17/09/2025 18:56:41	neelampandey99707@gmail.com	Aditi Pandey
20.	17/09/2025 19:08:56	fansarifarid253@gmail.com	Farid Ansari